

STAR Experiment Crosswalk v0-3

Current provenance assessment | Audit revision 0.3 | October 3, 2026

The current audit supplement follows. The complete supplied v0.2 PDF is retained afterward as the historical baseline. The current supplement governs audited evidence assessments where it differs from that baseline; it does not silently replace definitions or invent controlled experiments.

Authority: STAR Research Charter v0.2. Source locators and hashes: audit findings, data provenance records and code-log manifests in this package.

Current audit supplement - revision 0.3

Authority: supplied Research Charter v0.2. Audit date: October 3, 2026.

[Certain] Source inventory and static/table diagnostics are not independent scientific replication. This supplement governs current evidence assessment where it differs from the retained v0.2 baseline. Original claim wording, classifications, numerical reports and protocols remain historical and unchanged.

Coverage: 288 readable Drive snapshots; 14,361 local files inventoried. Semantic review prioritizes the linked evidence. Selected source hashes identify as-found bytes; readable Drive snapshot hashes do not establish canonical revision bytes. Large cloud-backed and excluded content is listed separately.

Evidence-state codes: U usable as properly labeled R&D; R reconstruction; N new test; C control; D derivation audit; H historical only; X negative/diagnostic. No record is promoted to independent, controlled, predictive or physical validation by this audit.

Current claim-to-evidence-to-experiment crosswalk (51)

All 43 original overall rows and eight historical assertions are explicitly accounted for. Original references are retained; audit relevance links do not confirm aliases or completed executions.

CORE-001 - Arithmetic-Cosmic Structural Correspondence

Original crosswalk references: EXP-CORE-A01.

Current registered protocol links: EXP-CTRL-A03, EXP-DATA-A01, EXP-DATA-A02, EXP-RANK-A01, EXP-TOPO-A01. Findings: E-LCL-005, E-LCL-006, E-LCL-014, E-LCL-016, E-DRV-004.

[Certain] Evidence state N; PROPOSED / CONTROLLED TEST NOT VERIFIED. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: EXP-CORE-A01 (candidate none; NO ALIAS ASSIGNED)

Next step: Record estimated/proxy status explicitly; prevent3-Selmer exactness or cosmic-evidence promotion from these filenames/results. Retain completed-run reports as historical computational evidence only; reconstruct identical mapping, filtration and survey/null comparisons before TOPO-001 promotion. Keep repository CSV namespace as legacy, preserve original files, and document explicit semantic crosswalk/conflicts before migrating IDs into the canonical update. Bind website displays to a synthetic/model provenance label; preserve model formulas and coefficients as construction choices. Keep COSMO-E002 downstream; displayed agreement with calibration anchors is not independent confirmation. Keep proposal, observed output and interpretation separate; require machine run manifests for execution claims.

RANK-001 - Rank/Environment Correspondence

Original crosswalk references: EXP-RANK-A01.

Current registered protocol links: EXP-RANK-A01. Findings: E-LCL-004, E-LCL-005.

[Certain] Evidence state N; PROPOSED / CONTROLLED TEST NOT VERIFIED. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: none

Next step: Retain partial lineage and missing-invariant diagnostics; bind exact upstream snapshot and preprocessing code to a registered dataset version before controlled testing. Record estimated/proxy status explicitly; prevent3-Selmer exactness or cosmic-evidence promotion from these filenames/results.

RANK-002 - Rank/Topological Complexity

Original crosswalk references: EXP-RANK-A02.

Current registered protocol links: EXP-RANK-A02, EXP-RANK-A01. Findings: E-LCL-007, E-LCL-008, E-LCL-009, E-LCL-010.

[Certain] Evidence state U + R + N; CANDIDATE/HISTORICAL EVIDENCE / RECONSTRUCTION REQUIRED. Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: none

Next step: Keep both generation IDs and exact input hashes; register RTCH-E1 as a separate local simulation/control if desired, without silently substituting it for canonical EXP-RTCH-B01. Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion. Exclude undefined-correlation p-values from support, preserve them as diagnostic failures, and require finite/constant-input checks before permutation reporting. Set Leakage Status POTENTIAL LEAKAGE for rank generalization and CONFIRMED TARGET CONTAMINATION for Faltings-held-out interpretation; rebuild fold-local embeddings and remove target-defining inputs for independent prediction tests.

RANK-003 - Rank-Stratified Cosmic Structure

Original crosswalk references: EXP-RANK-A03.

Current registered protocol links: EXP-RANK-A02, EXP-RANK-A01. Findings: E-LCL-005.

[Certain] Evidence state N; PROPOSED / CONTROLLED TEST NOT VERIFIED. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: EXP-RANK-A03 (candidate none; NO ALIAS ASSIGNED)

Next step: Record estimated/proxy status explicitly; prevent3-Selmer exactness or cosmic-evidence promotion from these filenames/results.

MAP-001 - Arithmetic Projection

Original crosswalk references: EXP-MAP-A01.

Current registered protocol links: EXP-MAP-A01. Findings: E-LCL-003, E-LCL-016.

[Certain] Evidence state U + R + N; CANDIDATE/HISTORICAL EVIDENCE / RECONSTRUCTION REQUIRED. Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: none

Next step: Preserve exact-path/hash revisions, flag copied manifests as inconsistent, and do not attribute working full-sample results to packaged subsets or blank outputs. Bind website displays to a synthetic/model provenance label; preserve model formulas and coefficients as construction choices. Keep COSMO-E002 downstream; displayed agreement with calibration anchors is not independent confirmation.

MAP-002 - Extended Projection

Original crosswalk references: EXP-MAP-A01.

Current registered protocol links: EXP-MAP-A01. Findings: E-LCL-003, E-LCL-004, E-LCL-016.

[Certain] Evidence state U + R + N; CANDIDATE/HISTORICAL EVIDENCE / RECONSTRUCTION REQUIRED.
Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: none

Next step: Preserve exact-path/hash revisions, flag copied manifests as inconsistent, and do not attribute working full-sample results to packaged subsets or blank outputs. Retain partial lineage and missing-invariant diagnostics; bind exact upstream snapshot and preprocessing code to a registered dataset version before controlled testing. Bind website displays to a synthetic/model provenance label; preserve model formulas and coefficients as construction choices. Keep COSMO-E002 downstream; displayed agreement with calibration anchors is not independent confirmation.

MAP-003 - Rank Elevation

Original crosswalk references: EXP-MAP-A02.

Current registered protocol links: EXP-MAP-A02, EXP-MAP-A01. Findings: E-LCL-003.

[Certain] Evidence state U + R + N; CANDIDATE/HISTORICAL EVIDENCE / RECONSTRUCTION REQUIRED.
Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: none

Next step: Preserve exact-path/hash revisions, flag copied manifests as inconsistent, and do not attribute working full-sample results to packaged subsets or blank outputs.

MAP-004 - Mapping Non-Bijectivity

Original crosswalk references: EXP-MAP-A01.

Current registered protocol links: EXP-MAP-A01. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state U + N; MODEL/METHODOLOGICAL REQUIREMENT RETAINED. Provenance: THEORETICAL / EXPLORATORY.

Unresolved links: none

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

MAP-005 - Alternative Mapping Families

Original crosswalk references: EXP-MAP-A01.

Current registered protocol links: EXP-MAP-A01. Findings: E-LCL-006.

[Certain] Evidence state U + R + N; CANDIDATE/HISTORICAL EVIDENCE / RECONSTRUCTION REQUIRED.
Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: none

Next step: Retain completed-run reports as historical computational evidence only; reconstruct identical mapping, filtration and survey/null comparisons before TOPO-001 promotion.

ARITH-001 - Fibonacci/Lucas Generators

Original crosswalk references: EXP-MAP-A03.

Current registered protocol links: EXP-MAP-A03, EXP-CTRL-A02. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state N; PROPOSED / CONTROLLED TEST NOT VERIFIED. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: none

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

TOPO-001 - Persistent Homology Correspondence

Original crosswalk references: EXP-TOPO-A01.

Current registered protocol links: EXP-TOPO-A01, EXP-RANK-A02. Findings: E-LCL-003, E-LCL-006, E-LCL-007, E-LCL-008, E-LCL-009, E-LCL-011.

[Certain] Evidence state U + R + N; CANDIDATE/HISTORICAL EVIDENCE / RECONSTRUCTION REQUIRED.
Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: none

Next step: Preserve exact-path/hash revisions, flag copied manifests as inconsistent, and do not attribute working full-sample results to packaged subsets or blank outputs. Retain completed-run reports as historical computational evidence only; reconstruct identical mapping, filtration and survey/null comparisons before TOPO-001 promotion. Keep both generation IDs and exact input hashes; register RTCH-E1 as a separate local simulation/control if desired, without silently substituting it for canonical EXP-RTCH-B01. Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion. Exclude undefined-correlation p-values from support, preserve them as diagnostic failures, and require finite/constant-input checks before permutation reporting. Record CSV stability output separately from empty JSON summary and require genuine perturbation/subsampling before claiming robustness.

TOPO-002 - Void Structure

Original crosswalk references: EXP-TOPO-A02.

Current registered protocol links: EXP-RANK-A02, EXP-TOPO-A01. Findings: E-LCL-006.

[Certain] Evidence state U + R + N; CANDIDATE/HISTORICAL EVIDENCE / RECONSTRUCTION REQUIRED.
Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: EXP-TOPO-A02 (candidate none; NO ALIAS ASSIGNED)

Next step: Retain completed-run reports as historical computational evidence only; reconstruct identical mapping, filtration and survey/null comparisons before TOPO-001 promotion.

TOPO-003 - Connected Components

Original crosswalk references: EXP-TOPO-A03.

Current registered protocol links: EXP-TOPO-A01. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state N; PROPOSED / CONTROLLED TEST NOT VERIFIED. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: EXP-TOPO-A03 (candidate none; NO ALIAS ASSIGNED)

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

MATH-K001 - Log-Normal Reciprocal Normalization

Original crosswalk references: EXP-K001.

Current registered protocol links: EXP-K001. Findings: E-DRV-001.

[Certain] Evidence state U + D + R; THEORETICAL STATEMENT / INDEPENDENT DERIVATION AUDIT REQUIRED. Provenance: THEORETICAL.

Unresolved links: none

Next step: Reconstruct assumptions, exact masses/units, median definition, seeds and sample selection. Retain unstable results and separate the expectation argument from physical interpretation.

MATH-K002 - Historical K Simulation

Original crosswalk references: EXP-K001.

Current registered protocol links: EXP-K001. Findings: E-DRV-001.

[Certain] Evidence state U + R; UNVERIFIED HISTORICAL RESULT / RECONSTRUCTION REQUIRED. Provenance: SIMULATION / HISTORICAL MODEL.

Unresolved links: none

Next step: Reconstruct assumptions, exact masses/units, median definition, seeds and sample selection. Retain unstable results and separate the expectation argument from physical interpretation.

MATH-K003 - Small-Sample K Instability

Original crosswalk references: EXP-K002.

Current registered protocol links: No canonical link substituted. Findings: E-DRV-001.

[Certain] Evidence state U + X + R; NEGATIVE HISTORICAL REPORT RETAINED / NOT RERUN. Provenance: HISTORICAL MODEL / NEGATIVE/NULL.

Unresolved links: EXP-K002 (candidate none; NO ALIAS ASSIGNED)

Next step: Reconstruct assumptions, exact masses/units, median definition, seeds and sample selection. Retain unstable results and separate the expectation argument from physical interpretation.

COSMO-L001 - Phenomenological L-cosmo

Original crosswalk references: EXP-L001.

Current registered protocol links: EXP-L001, EXP-CTRL-A03. Findings: E-LCL-012, E-DRV-001, E-DRV-003.

[Certain] Evidence state U + R + N; CANDIDATE/HISTORICAL EVIDENCE / RECONSTRUCTION REQUIRED.
Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: none

Next step: Retain M/H categories and R evidence state; set preprocessing leakage concern, rename/cross-reference cosmo_rank semantically, bind the actual input revision, and conduct fold-local incremental models before promotion. Reconstruct assumptions, exact masses/units, median definition, seeds and sample selection. Retain unstable results and separate the expectation argument from physical interpretation. Retain proposal/code provenance; validate an exact runnable variant, training-fold derived quantities, outcome definition and locked independent test.

COSMO-L002 - L-cosmo and SFR

Original crosswalk references: EXP-SFR-A01.

Current registered protocol links: EXP-SFR-A01. Findings: E-LCL-012, E-DRV-002, E-DRV-003.

[Certain] Evidence state U + R + N; CANDIDATE/HISTORICAL EVIDENCE / RECONSTRUCTION REQUIRED.
Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: none

Next step: Retain M/H categories and R evidence state; set preprocessing leakage concern, rename/cross-reference cosmo_rank semantically, bind the actual input revision, and conduct fold-local incremental models before promotion. Bind a precise executed code/data revision and reconstruct fold-local preprocessing before crediting reported predictive performance. Retain proposal/code provenance; validate an exact runnable variant, training-fold derived quantities, outcome definition and locked independent test.

COSMO-R001 - Morphology/Redshift Composite Feature

Original crosswalk references: CTRL-COSMO-001.

Current registered protocol links: No canonical link substituted. Findings: E-LCL-012.

[Certain] Evidence state U + H; CONFIRMED ENGINEERED FEATURE IN INSPECTED SCRIPT. Provenance: HISTORICAL MODEL.

Unresolved links: CTRL-COSMO-001 (candidate none; NO ALIAS ASSIGNED)

Next step: Retain M/H categories and R evidence state; set preprocessing leakage concern, rename/cross-reference cosmo_rank semantically, bind the actual input revision, and conduct fold-local incremental models before promotion.

SFR-A001 - Arithmetic Incremental SFR Information

Original crosswalk references: EXP-SFR-A01.

Current registered protocol links: EXP-SFR-A01, EXP-CTRL-A03, EXP-CTRL-A02. Findings: E-LCL-012, E-LCL-013, E-DRV-002, E-DRV-003.

[Certain] Evidence state U + R + N; CANDIDATE/HISTORICAL EVIDENCE / RECONSTRUCTION REQUIRED.
Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: none

Next step: Retain M/H categories and R evidence state; set preprocessing leakage concern, rename/cross-reference cosmo_rank semantically, bind the actual input revision, and conduct fold-local incremental models before promotion. Classify as historical random-feature stellar-mass baseline/control candidate, preserve absent-target diagnostic, and require exact input/schema plus explicit outcome exclusion and SFR-specific control reconstruction. Bind a precise executed code/data revision and reconstruct fold-local preprocessing before crediting reported predictive performance. Retain proposal/code provenance; validate an exact runnable variant, training-fold derived quantities, outcome definition and locked independent test.

SFR-A002 - Arithmetic-Only SFR Prediction

Original crosswalk references: EXP-SFR-A02.

Current registered protocol links: EXP-SFR-A02. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state N; PROPOSED / CONTROLLED TEST NOT VERIFIED. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: none

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

SFR-E001 - Extreme-SFR Model Behavior

Original crosswalk references: EXP-SFR-A03.

Current registered protocol links: EXP-SFR-A03. Findings: E-DRV-002.

[Certain] Evidence state U + R; UNVERIFIED HISTORICAL RESULT / RECONSTRUCTION REQUIRED. Provenance: HISTORICAL MODEL.

Unresolved links: none

Next step: Bind a precise executed code/data revision and reconstruct fold-local preprocessing before crediting reported predictive performance.

SFR-A003 - Extreme-SFR Robustness

Original crosswalk references: EXP-SFR-A03.

Current registered protocol links: EXP-SFR-A03. Findings: E-DRV-002.

[Certain] Evidence state U + N; MODEL/METHODOLOGICAL REQUIREMENT RETAINED. Provenance: THEORETICAL / EXPLORATORY.

Unresolved links: none

Next step: Bind a precise executed code/data revision and reconstruct fold-local preprocessing before crediting reported predictive performance.

CTRL-RF001 - Random Feature Baseline

Original crosswalk references: EXP-CTRL-A01.

Current registered protocol links: EXP-CTRL-A01. Findings: E-LCL-013, E-LCL-014.

[Certain] Evidence state U + C + X + R; HISTORICAL CONTROL CANDIDATE / INPUT SCHEMA MISMATCH.
Provenance: HISTORICAL MODEL.

Unresolved links: none

Next step: Classify as historical random-feature stellar-mass baseline/control candidate, preserve absent-target diagnostic, and require exact input/schema plus explicit outcome exclusion and SFR-specific control reconstruction. Keep repository CSV namespace as legacy, preserve original files, and document explicit semantic crosswalk/conflicts before migrating IDs into the canonical update.

CTRL-A001 - Arithmetic Permutation Control

Original crosswalk references: EXP-CTRL-A02.

Current registered protocol links: EXP-CTRL-A02. Findings: E-LCL-009, E-LCL-014.

[Certain] Evidence state N + X; CONTROL REQUIRED / INVALID LOCAL SIGNIFICANCE RETAINED. Provenance: NEGATIVE/NULL / EXPLORATORY.

Unresolved links: none

Next step: Exclude undefined-correlation p-values from support, preserve them as diagnostic failures, and require finite/constant-input checks before permutation reporting. Keep repository CSV namespace as legacy, preserve original files, and document explicit semantic crosswalk/conflicts before migrating IDs into the canonical update.

CTRL-A002 - Leakage-Controlled Reconstruction

Original crosswalk references: EXP-CTRL-A03.

Current registered protocol links: EXP-CTRL-A03. Findings: E-LCL-012, E-LCL-014, E-DRV-002, E-DRV-003.

[Certain] Evidence state X + R; STATIC LEAKAGE ROUTES DIAGNOSED / CLEAN RERUN REQUIRED. Provenance: HISTORICAL MODEL / NEGATIVE/NULL.

Unresolved links: none

Next step: Retain M/H categories and R evidence state; set preprocessing leakage concern, rename/cross-reference cosmo_rank semantically, bind the actual input revision, and conduct fold-local incremental models before promotion. Keep repository CSV namespace as legacy, preserve original files, and document explicit semantic crosswalk/conflicts before migrating IDs into the canonical update. Bind a precise executed code/data revision and reconstruct fold-local preprocessing before crediting reported predictive performance. Retain proposal/code provenance; validate an exact runnable variant, training-fold derived quantities, outcome definition and locked independent test.

DATA-M001 - Coordinate Matching

Original crosswalk references: EXP-DATA-A01.

Current registered protocol links: EXP-DATA-A01. Findings: E-LCL-001, E-LCL-002.

[Certain] Evidence state X + R; FAILED AS-FOUND ASSOCIATIONS / RECONSTRUCTION REQUIRED. Provenance: HISTORICAL MODEL / NEGATIVE/NULL.

Unresolved links: none

Next step: Retain failed data-quality evidence; quarantine these tables from controlled analyses until matching is rebuilt from source catalogs with spherical distance, keyed joins, duplicate controls, tolerance sensitivity, and saved data/code hashes. Record code variants separately and reconstruct EXP-DATA-A01 using a registered coordinate system, exact input IDs, spherical matching, and explicit identity crosswalks.

DATA-M002 - Multi-Survey Integration

Original crosswalk references: DATA-PROV-001.

Current registered protocol links: EXP-DATA-A02. Findings: E-LCL-001, E-LCL-002, E-LCL-003, E-LCL-004, E-LCL-014, E-LCL-015, E-DRV-004.

[Certain] Evidence state U + X + R; PARTIAL LINEAGE / ARTIFACT CONFLICTS. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: DATA-PROV-001 (candidate EXP-DATA-A02; UNCONFIRMED)

Next step: Retain failed data-quality evidence; quarantine these tables from controlled analyses until matching is rebuilt from source catalogs with spherical distance, keyed joins, duplicate controls, tolerance sensitivity, and saved data/code hashes. Record code variants separately and reconstruct EXP-DATA-A01 using a registered coordinate system, exact input IDs, spherical matching, and explicit identity crosswalks. Preserve exact-path/hash revisions, flag copied manifests as inconsistent, and do not attribute working full-sample results to packaged subsets or blank outputs. Retain partial lineage and missing-invariant diagnostics; bind exact upstream snapshot and preprocessing code to a registered dataset version before controlled testing. Keep repository CSV namespace as legacy, preserve original files, and document explicit semantic crosswalk/conflicts before migrating IDs into the canonical update. Preserve upstream identifiers as observed provenance fields; add exact release/query and processing/code bindings before declaring complete lineage. Keep proposal, observed output and interpretation separate; require machine run manifests for execution claims.

RTCH-M001 - Variational RTCH Construction

Original crosswalk references: EXP-RTCH-B01.

Current registered protocol links: EXP-RTCH-B01. Findings: E-LCL-007, E-LCL-008, E-LCL-010, E-LCL-011.

[Certain] Evidence state U + R + N; CANDIDATE/HISTORICAL EVIDENCE / RECONSTRUCTION REQUIRED. Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: none

Next step: Keep both generation IDs and exact input hashes; register RTCH-E1 as a separate local simulation/control if desired, without silently substituting it for canonical EXP-RTCH-B01. Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion. Set Leakage Status POTENTIAL LEAKAGE for rank generalization and CONFIRMED TARGET CONTAMINATION for Faltings-held-out interpretation; rebuild fold-local embeddings and remove target-defining inputs for independent prediction tests. Record CSV stability output separately from empty JSON summary and require genuine perturbation/subsampling before claiming robustness.

RTCH-M002 - Closed Pullback Two-Form

Original crosswalk references: EXP-RTCH-B01.

Current registered protocol links: EXP-RTCH-B01. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state U + D + R; THEORETICAL STATEMENT / INDEPENDENT DERIVATION AUDIT REQUIRED. Provenance: THEORETICAL.

Unresolved links: none

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

RTCH-H001 - Topological Coupling

Original crosswalk references: EXP-RTCH-B02.

Current registered protocol links: EXP-RTCH-B02. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state N; PROPOSED / CONTROLLED TEST NOT VERIFIED. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: none

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

RTCH-H002 - Conformal Trace Coupling

Original crosswalk references: EXP-RTCH-B03.

Current registered protocol links: EXP-RTCH-B03. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state U + D + R; THEORETICAL STATEMENT / INDEPENDENT DERIVATION AUDIT REQUIRED. Provenance: THEORETICAL.

Unresolved links: none

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

RTCH-H003 - Fifth-Force Acceleration

Original crosswalk references: EXP-RTCH-B02.

Current registered protocol links: EXP-RTCH-B02. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state U + D + R; THEORETICAL STATEMENT / INDEPENDENT DERIVATION AUDIT REQUIRED. Provenance: THEORETICAL.

Unresolved links: none

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

RTCH-H004 - Unified Matter Description

Original crosswalk references: EXP-RTCH-B01.

Current registered protocol links: EXP-RTCH-B01. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state U + N; MODEL/METHODOLOGICAL REQUIREMENT RETAINED. Provenance: THEORETICAL / EXPLORATORY.

Unresolved links: none

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

FIELD-001 - Arithmetic Topography

Original crosswalk references: EXP-FIELD-B01.

Current registered protocol links: EXP-FIELD-B01. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state U + N; MODEL/METHODOLOGICAL REQUIREMENT RETAINED. Provenance: THEORETICAL / EXPLORATORY.

Unresolved links: none

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

FIELD-002 - Arithmetic Field as Gravity

Original crosswalk references: EXP-FIELD-B02.

Current registered protocol links: EXP-FIELD-B02. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state N; PROPOSED / CONTROLLED TEST NOT VERIFIED. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: none

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

FIELD-003 - Arithmetic Tensor / Einstein Equation Extension

Original crosswalk references: EXP-FIELD-B03.

Current registered protocol links: EXP-FIELD-B02. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state N; PROPOSED / CONTROLLED TEST NOT VERIFIED. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: EXP-FIELD-B03 (candidate none; NO ALIAS ASSIGNED)

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

ECC-001 - Entropy/Cohomology Structure

Original crosswalk references: EXP-ECC-B01.

Current registered protocol links: EXP-ECC-B01. Findings: E-LCL-007, E-LCL-008, E-LCL-010, E-LCL-011, E-DRV-004.

[Certain] Evidence state U + R + N; CANDIDATE/HISTORICAL EVIDENCE / RECONSTRUCTION REQUIRED. Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: none

Next step: Keep both generation IDs and exact input hashes; register RTCH-E1 as a separate local simulation/control if desired, without silently substituting it for canonical EXP-RTCH-B01. Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion. Set Leakage Status POTENTIAL LEAKAGE for rank generalization and CONFIRMED TARGET CONTAMINATION for Faltings-held-out interpretation; rebuild fold-local embeddings and remove target-defining inputs for independent prediction tests. Record CSV stability output separately from empty JSON summary and require genuine perturbation/subsampling before claiming robustness. Keep proposal,

observed output and interpretation separate; require machine run manifests for execution claims.

ECC-002 - Closed Non-Exact Requirement

Original crosswalk references: EXP-ECC-B01.

Current registered protocol links: EXP-ECC-B01. Findings: E-LCL-007.

[Certain] Evidence state U + D + R; THEORETICAL STATEMENT / INDEPENDENT DERIVATION AUDIT REQUIRED.
Provenance: THEORETICAL.

Unresolved links: none

Next step: Keep both generation IDs and exact input hashes; register RTCH-E1 as a separate local simulation/control if desired, without silently substituting it for canonical EXP-RTCH-B01.

ECC-003 - Entropy-Energy Relation

Original crosswalk references: EXP-ECC-B02.

Current registered protocol links: EXP-ENERGY-B01. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state N; PROPOSED / CONTROLLED TEST NOT VERIFIED. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: EXP-ECC-B02 (candidate none; NO ALIAS ASSIGNED)

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

ENERGY-001 - Generalized Mass-Energy Relation

Original crosswalk references: EXP-ENERGY-B01.

Current registered protocol links: EXP-ENERGY-B01. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state N; PROPOSED / CONTROLLED TEST NOT VERIFIED. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: none

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

COSMO-E001 - Arithmetic/Topological Redshift Evolution

Original crosswalk references: EXP-COSMO-B01.

Current registered protocol links: EXP-COSMO-B01. Findings: Original conceptual/methodological provenance only.

[Certain] Evidence state N; PROPOSED / CONTROLLED TEST NOT VERIFIED. Provenance: EXPLORATORY / UNVERIFIED.

Unresolved links: none

Next step: Follow the original registered test/derivation and charter controls; bind source releases, exact code, parameters, seeds, nulls and held-out results before promotion.

COSMO-E002 - Hubble-Planck Tension Application

Original crosswalk references: EXP-COSMO-B02.

Current registered protocol links: EXP-COSMO-C01. Findings: E-LCL-016.

[Certain] Evidence state N + H; DOWNSTREAM / UNVALIDATED; ILLUSTRATION IS CALIBRATED. Provenance: EXPLORATORY / HISTORICAL MODEL.

Unresolved links: EXP-COSMO-B02 (candidate EXP-COSMO-C01; UNCONFIRMED)

Next step: Bind website displays to a synthetic/model provenance label; preserve model formulas and coefficients as construction choices. Keep COSMO-E002 downstream; displayed agreement with calibration anchors is not independent confirmation.

HIST-001 - Retained historical assertion 001

Original crosswalk references: No individual row in original Overall Crosswalk; retained historical assertion.

Current registered protocol links: EXP-RANK-A01, EXP-TOPO-A01, EXP-MAP-A01. Findings: E-LCL-006, E-LCL-008, E-LCL-016.

[Certain] Evidence state H + R; HISTORICAL ONLY / RECONSTRUCTION AND NEW CONTROLLED TEST REQUIRED. Provenance: HISTORICAL MODEL / UNVERIFIED.

Unresolved links: none

Next step: Retain completed-run reports as historical computational evidence only; reconstruct identical mapping, filtration and survey/null comparisons before TOPO-001 promotion. Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion. Bind website displays to a synthetic/model provenance label; preserve model formulas and coefficients as construction choices. Keep COSMO-E002 downstream; displayed agreement with calibration anchors is not independent confirmation.

HIST-002 - Retained historical assertion 002

Original crosswalk references: No individual row in original Overall Crosswalk; retained historical assertion.

Current registered protocol links: EXP-RANK-A01, EXP-RANK-A02. Findings: E-LCL-005, E-LCL-008.

[Certain] Evidence state H + R; HISTORICAL ONLY / RECONSTRUCTION AND NEW CONTROLLED TEST REQUIRED. Provenance: HISTORICAL MODEL / UNVERIFIED.

Unresolved links: none

Next step: Record estimated/proxy status explicitly; prevent3-Selmer exactness or cosmic-evidence promotion from these filenames/results. Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion.

HIST-003 - Retained historical assertion 003

Original crosswalk references: No individual row in original Overall Crosswalk; retained historical assertion.

Current registered protocol links: EXP-RANK-A01, EXP-RANK-A02. Findings: E-LCL-005, E-LCL-008.

[Certain] Evidence state H + R; HISTORICAL ONLY / RECONSTRUCTION AND NEW CONTROLLED TEST REQUIRED. Provenance: HISTORICAL MODEL / UNVERIFIED.

Unresolved links: none

Next step: Record estimated/proxy status explicitly; prevent3-Selmer exactness or cosmic-evidence promotion from these filenames/results. Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion.

HIST-004 - Retained historical assertion 004

Original crosswalk references: No individual row in original Overall Crosswalk; retained historical assertion.

Current registered protocol links: EXP-RANK-A01, EXP-RANK-A02. Findings: E-LCL-005, E-LCL-008.

[Certain] Evidence state H + R; HISTORICAL ONLY / RECONSTRUCTION AND NEW CONTROLLED TEST REQUIRED. Provenance: HISTORICAL MODEL / UNVERIFIED.

Unresolved links: none

Next step: Record estimated/proxy status explicitly; prevent3-Selmer exactness or cosmic-evidence promotion from these filenames/results. Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion.

HIST-005 - Retained historical assertion 005

Original crosswalk references: No individual row in original Overall Crosswalk; retained historical assertion.

Current registered protocol links: EXP-RANK-A01, EXP-RANK-A02. Findings: E-LCL-005, E-LCL-008.

[Certain] Evidence state H + R; HISTORICAL ONLY / RECONSTRUCTION AND NEW CONTROLLED TEST REQUIRED. Provenance: HISTORICAL MODEL / UNVERIFIED.

Unresolved links: none

Next step: Record estimated/proxy status explicitly; prevent3-Selmer exactness or cosmic-evidence promotion from these filenames/results. Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion.

HIST-006 - Retained historical assertion 006

Original crosswalk references: No individual row in original Overall Crosswalk; retained historical assertion.

Current registered protocol links: EXP-RANK-A01, EXP-RANK-A02. Findings: E-LCL-005, E-LCL-008.

[Certain] Evidence state H + R; HISTORICAL ONLY / RECONSTRUCTION AND NEW CONTROLLED TEST REQUIRED. Provenance: HISTORICAL MODEL / UNVERIFIED.

Unresolved links: none

Next step: Record estimated/proxy status explicitly; prevent3-Selmer exactness or cosmic-evidence promotion from these filenames/results. Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion.

HIST-007 - Retained historical assertion 007

Original crosswalk references: No individual row in original Overall Crosswalk; retained historical assertion.

Current registered protocol links: EXP-TOPO-A01. Findings: E-LCL-006, E-LCL-008, E-LCL-016.

[Certain] Evidence state H + R; HISTORICAL ONLY / RECONSTRUCTION AND NEW CONTROLLED TEST REQUIRED. Provenance: HISTORICAL MODEL / UNVERIFIED.

Unresolved links: none

Next step: Retain completed-run reports as historical computational evidence only; reconstruct identical mapping, filtration and survey/null comparisons before TOPO-001 promotion. Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion. Bind website displays to a synthetic/model provenance label; preserve model formulas and coefficients as construction choices. Keep COSMO-E002 downstream; displayed agreement with calibration anchors is not independent confirmation.

HIST-008 - Retained historical assertion 008

Original crosswalk references: No individual row in original Overall Crosswalk; retained historical assertion.

Current registered protocol links: EXP-DATA-A02, EXP-TOPO-A01, EXP-MAP-A01. Findings: E-LCL-006, E-LCL-008, E-LCL-016.

[Certain] Evidence state H + R; HISTORICAL ONLY / RECONSTRUCTION AND NEW CONTROLLED TEST REQUIRED. Provenance: HISTORICAL MODEL / UNVERIFIED.

Unresolved links: none

Next step: Retain completed-run reports as historical computational evidence only; reconstruct identical mapping, filtration and survey/null comparisons before TOPO-001 promotion. Retain both null/diagnostic generations. Rebuild adequate null counts, effect sizes, intervals, multiple-comparison controls, code binding and independent arithmetic/astronomical inputs before empirical promotion. Bind website displays to a synthetic/model provenance label; preserve model formulas and coefficients as construction choices. Keep COSMO-E002 downstream; displayed agreement with calibration anchors is not independent confirmation.

Source locators for new findings

Exact source file paths, SHA256 digests, line/table locators and interpretation limits are in local_findings.json and drive_findings.json. Evidence presence is distinct from validation.

E-LCL-001 - Coordinate-match artifacts fail their claimed tolerance

[Certain] Both integrated SDSS/HI tables contain 1,495 rows linked to the same MANGAID 1-48157. Every recomputed great-circle separation exceeds 2 arcsec (minimum 741.9258354463, maximum 445141.8800654686). The with_arcsec table stores inf in all 1,495 separation rows; the other table stores only values above 2 arcsec. These artifacts do not substantiate a valid ≤ 2 arcsec match.

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\merged_sdss_hi.csv", "sha256":  
"5a9606b75bc55de5e7833a04e5c56318df9f21a31560df180db2707374a22d50", "locator": "CSV lines 1\u20131496; line  
2 is an explicit counterexample", "observation": "1495 rows; 1494 duplicate MANGAID excess; first actual spherical  
separation 276891.4500653436 arcsec"}
```

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\merged_sdss_hi_with_arcsec.csv", "sha256":  
"e6e2e749d0ec45b5713311dfe3b3cc958fdbb13166f770ce2c47a30dbd5f8dad", "locator": "CSV lines 1\u20131496;  
arcsec_separation column", "observation": "1495 nonfinite stored separations"}
```

```
{"path": "C:\\Users\\pmqr7\\codex\\.chatgpt-projects\\lg-p-6aa87346c5e88191979368ee9ee5ce5b\\audit_work\\local_ma  
tch_checks.json", "sha256": "c1dc53be08a6580e88ebdc0ebf884d5184f4fb8318763f38ebd1077fde2c987", "locator":  
"complete independent table diagnostic", "observation": ""}
```

E-LCL-002 - Matched-catalog code lacks fully controlled spherical/keyed lineage

[Certain] A 2 arcsec matching candidate exists, but uses Euclidean distance on (RA_rad,Dec_rad), rather than great-circle angular distance. A later integration script relabels MagPhys CATAID as objid without an explicit identifier crosswalk. Other STAR script generations use 10 or 30 arcsec, so there is no single demonstrated locked match protocol.

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\pip.py", "sha256":  
"912eca01f5bee760f5075560460d4928bc842446cddec0281a8e713bfe7e8a91", "locator": "lines16\\u201321  
and24\\u201335", "observation": "two-dimensional radian KDTree, cutoff2arcsec, finite-filtered matches, output  
merged_sdssdr18_mangaHI_kdtree.csv"}
```

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\singularity.py", "sha256":  
"0ed13876725511c846b3c94fdc21db8636298297163cb6c1ceaae76e9b0b3283", "locator": "lines20\\u201324  
and35\\u201343", "observation": "CATAID relabeled as objid then joined"}
```

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\STAR.py", "sha256":  
"df10464ffb6bee489d78d52e75a515ecfe179e2a27c60dca679e74cd6b966ef3", "locator":  
"lines95,110\\u2013115,183\\u2013195,214,222,230", "observation": "10arcsec default and matched_df1=df1.copy after  
collecting chunk matches"}
```

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\STAR5.py", "sha256":  
"d44de89fa69032b5ca8711a2a1c143656a423a2cebb9a41fb1e57a6768329f2b", "locator":  
"lines42,52\\u2013135,97,103,107", "observation": "30arcsec variant with index merges"}
```

E-LCL-003 - Packaged projection manifest disagrees with adjacent artifact

[Certain] S.T.A.R.-Program/data/derived/acsc_projected_cremona_final.csv is 2 bytes with no data/header, but its adjacent manifest reports 38,042 output rows. Both packaged raw arithmetic datasets contain 1,500 rows. The working kepler/derived/acsc_projected_cremona_final.csv contains 38,042 rows. Repeated filenames refer to materially different revisions.

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\data\\derived\\acsc_projected_cremona_final.csv",  
"sha256": "7eb70257593da06f682a3ddda54a9d260d4fc514f645237f5ca74b08f8da61a6", "locator": "entire2byte file",  
"observation": "no data/header"}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\data\\derived\\acsc_projected_cremona_final.csv.m  
anifest.json", "sha256": "5980d29db9705eba87643f749f443da1d4f779430cfd5ff4e61bb4b0655ba171", "locator": "JSON  
rows_output", "observation": "38042"}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\data\\raw\\cremona_raw_parsed.csv", "sha256":  
"960a96877035b51d8dd22bee229a00ef512442782d07f7f0529c92c33f00cbc0", "locator": "CSV1500data rows",  
"observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\data\\raw\\lmfdb_raw_parsed.csv", "sha256":  
"8f9ba5410bdbcc980dce7c915af5ca35b4c1dfce0c08229688b5b38308d1f607", "locator": "CSV1500data rows",  
"observation": ""}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\derived\\acsc_projected_cremona_final.csv", "sha256":  
"ca37c71b32151ee069b427a1cddc2f446bbe6395c813ddac27a02d815ff77bc2", "locator": "CSV38042data rows",  
"observation": ""}
```

E-LCL-004 - Arithmetic validation manifests omit several required invariants

[Certain] Root Cremona and LMFDB validation manifests name raw inputs and output tables and declare 38,042/3,064,705 rows. Both declare regulator, real_period and torsion_order absent for every row. They do not record input/output SHA256, exact upstream release, code revision, environment, or experiment ID.

```
{
  "path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\acsc_validation_cremona.manifest.json",
  "sha256": "ab40c566b08a12339887ab06f91b7969bea812da06abcb3f96646d8d1e1ecebe",
  "locator": "JSON input_file,output_file,rows_input,rows_output,diagnostics.null_counts",
  "observation": ""
}

{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\acsc_validation_lmfdb.manifest.json", "sha256": "9103710b256c42bf182c2e4e0f6ddd1c8cc1e14b206f22263b3c51c1a3de1a3a", "locator": "JSON input_file,output_file,rows_input,rows_output,diagnostics.null_counts", "observation": ""}

{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\cremona_raw_parsed.csv", "sha256": "135f01c3e07390b3fba0784cf91b8c0e1799562e77ad670923c05683e8563e11", "locator": "CSVheader and full checked table", "observation": ""}

{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\lmfdb_raw_parsed.csv", "sha256": "bdb6bd1bd5b9fac8d7210623dbd2017962b3a75963c6581cda2f431b6f2d41b0", "locator": "CSVheader and full checked table", "observation": ""}
```

E-LCL-005 - Estimated3-Selmer quantities are explicitly proxies

[Certain] Arithmetic enrichment artifacts label the relevant field `estimated_3_selmer_bound`, alongside `sage_rank`, `pari_2_selmer_rank` and `pari_analytic_rank`. The name and schema distinguish an estimate from a computed3-Selmer group; no astronomical correspondence follows from this arithmetic calculation.

```
{
  "path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\acsc_final_combined.csv",
  "sha256": "b2248d76c9c274d0fea23f3c84b7cb54519c9b33cd48c4d857bed53e30d50bb4",
  "locator": "CSVheader and initial rows; estimated_3_selmer_bound column",
  "observation": ""
}

{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\acsc_final_cremona.csv", "sha256": "4a865af2577e5f98491de35d82c398a7e5cad6bf8ecaad5f7ce1fe56401471df", "locator": "CSVheader", "observation": ""}

{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\acsc_final_lmfdb.csv", "sha256": "98b9f1b3c1e9e2eb3c36729c3e5599385f3d32e1f49f3301743a9c170f38c97c", "locator": "CSVheader", "observation": ""}
```

E-LCL-006 - PTD batch completion is computational lineage without cosmic/null validation

[Certain] The root PTD aggregate manifest reports 127 chunks, 127 completed, 0 failed, chunk 300, maxdim1 and a numerical filtration threshold. It names arithmetic coordinates and pickle outputs, but not independent cosmic input, null ensemble, effect uncertainty, mapped Claim ID or canonical experiment ID. maxdim1 cannot supply H2 void topology.

```
{
  "path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\derived\\tda_ptd_batches\\aggregate_manifest.json",
  "sha256": "21114fe5cbbd7fd58bc3baeb512fab1161289f89f2c162ff0f1bad41c255f004",
  "locator": "JSON n_chunks,completed,failed,params and result_files",
  "observation": ""
}

{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\derived\\tda_ptd_batches\\summary_per_chunk.json", "sha256": "55322e632b7c6198adb38676df693b7c08b2dee353825eb29692985cb90fe298", "locator": "per-chunk result metadata", "observation": ""}
```

E-LCL-007 - Two RTCH-E1 generations are synthetic controls with matching input hashes

[Certain] Outer results are a 900-point demo recorded 2026-09-18T20:42:17Z; nested results are a 350-point demo recorded 2026-09-18T21:48:45Z. Both declared input SHA256 values match the corresponding as-found demo_torus.csv. The scripts differ: root old implementation sends sparse kNN adjacency to `riper(distance_matrix=False)`; current program version describes/uses complete Euclidean VR and limits demo 350.

Neither manifest binds a code revision/hash or full package versions.

```
{
  "path": "\\\wsl.localhost\Ubuntu\home\kepler\S.T.A.R.-Program\RTCH_E1\results\run_manifest.json",
  "sha256": "331fc7e3c749ff1f0c12632e1d77ba73096ffe6f55e6e40bc3a0e5fe10978925",
  "locator": "JSON input_sha256,n_analysis,config,software",
  "observation": ""
}

{
  "path": "\\\wsl.localhost\Ubuntu\home\kepler\S.T.A.R.-Program\RTCH_E1\data\demo_torus.csv",
  "sha256": "2790c3bdebb665ffa9a5df3ec6b8093dc678c743303b56732be751678bdcfce7",
  "locator": "entire CSV;900data rows",
  "observation": ""
}

{
  "path": "\\\wsl.localhost\Ubuntu\home\kepler\S.T.A.R.-Program\RTCH_E1\RTCH_E1\results\run_manifest.json",
  "sha256": "bd3c605cdcf76c7ba0d348b3266024165c9133e19f4b3b81fb82d81a8b2a5901",
  "locator": "JSON input_sha256,demo_mode,n_analysis,topology_method",
  "observation": ""
}

{
  "path": "\\\wsl.localhost\Ubuntu\home\kepler\S.T.A.R.-Program\RTCH_E1\RTCH_E1\data\demo_torus.csv",
  "sha256": "ce3e2d8dd919c83362de1fcbe099f8074abe8a693d1dc2664da6956240021e69",
  "locator": "entire CSV;350data rows",
  "observation": ""
}

{
  "path": "\\\wsl.localhost\Ubuntu\home\kepler\rtch_e1.py",
  "sha256": "37b0763b7e2677ea080ef390588fa65b1486bb5438d0e8e72658641401e888f4",
  "locator": "lines334,862\u201313885",
  "observation": "old sparse-adjacency method; n=900 synthetic generator"
}

{
  "path": "\\\wsl.localhost\Ubuntu\home\kepler\S.T.A.R.-Program\RTCH_E1\rtch_e1.py",
  "sha256": "4fe93acda593d564ab204efac217cdf9aa76c256db57d057aa17ef29dc75a1f",
  "locator": "lines371\u201313407,985\u2013131011",
  "observation": "corrected completeVR; n=350 synthetic generator"
}
```

E-LCL-008 - RTCH demo results do not pass declared null criterion

[Certain] The 900-point result reports all topology-null empirical p-values 1.0 and no H1/H2/H3 summary. The 350-point result has H1max persistence 0.7075445652 and H2max 0.3834588528, but null p-values 1/3 with only 2 nulls at alpha 0.01; minimum attainable plus-one p is 1/3, so the declared significance gate cannot be met. The nested graph-eccentricity rank $cvR2=0.17245548$ is a synthetic-demo metric, not controlled astronomical prediction.

```
{
  "path": "\\\wsl.localhost\Ubuntu\home\kepler\S.T.A.R.-Program\RTCH_E1\results\result.json",
  "sha256": "09136ef0b8d380f09f49f968d9f749673a2470bb8557353e4d31566dddf94460",
  "locator": "JSON topology and nulls",
  "observation": ""
}

{
  "path": "\\\wsl.localhost\Ubuntu\home\kepler\S.T.A.R.-Program\RTCH_E1\RTCH_E1\results\result.json",
  "sha256": "d294a3426d69f565d33b07b641768479d04bc12b586cb2b2f17e9f9d9b64cef4",
  "locator": "JSON topology.intrinsic,H1/H2 nulls",
  "observation": ""
}

{
  "path": "\\\wsl.localhost\Ubuntu\home\kepler\S.T.A.R.-Program\RTCH_E1\RTCH_E1\results\prediction_tests.csv",
  "sha256": "30b91aed6a6a687b2fe73cdebed209552e6f41f1af1127e1110e9bc7ca26f3e6",
  "locator": "CSVline5 rank/graph_eccentricity",
  "observation": ""
}

{
  "path": "\\\wsl.localhost\Ubuntu\home\kepler\S.T.A.R.-Program\RTCH_E1\RTCH_E1\results\run_manifest.json",
  "sha256": "bd3c605cdcf76c7ba0d348b3266024165c9133e19f4b3b81fb82d81a8b2a5901",
  "locator": "JSON n_nulls=2,alpha=.01",
  "observation": ""
}
```

E-LCL-009 - RTCH permutation formula can yield false significance from NaN

[Certain] Outer rank_label_nulls rows pc2/pc3 have blank observed/null Spearman diagnostics yet permutation $p=0.009900990099009901$. Both outer scalar columns are constant 0 across all 900 rows (min=max=0). Static code does not check finite observed/null correlations before $(1+\text{sum}(\text{vals}\geq\text{observed})) / (1+\text{len}(\text{vals}))$. If observed is NaN, all comparisons are False and the formula returns 1/101; that number cannot support significance.

```
{
  "path": "\\\wsl.localhost\Ubuntu\home\kepler\S.T.A.R.-Program\RTCH_E1\results\rank_label_nulls.csv",
  "sha256": "d8b357e6a10b52279605f26a5b6f233f2eb2e954edc9f6d744185523fbf3ad32",
  "locator": "CSVlines3\u2013134",
  "observation": ""
}
```


[Certain] L-cosmo(s).py performs global median imputation, MinMax fit, massmedian, redshiftbinning and groupmeans before train/test splitting. cosmo_rank is a weighted morphology/confidence/redshift feature, not elliptic-curve rank. L_cosmo_s* columns are constructed/plotted but omitted from the model feature list; this script therefore cannot establish incremental predictive value for L_cosmo. Its groupmean Schechter form differs from the canonical mass-ratio sum.

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\L-cosmo(s).py", "sha256":  
"408dca936a416be0f25d1dcc705e302c58a72cfbfb5661441c6297872d5df297", "locator":  
"lines22,35\\u201344,50\\u201361,97\\u2013105", "observation": "global transformations and no L_cosmo_s* in  
modelfeatures"}  
  
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\L-cosmos.py", "sha256":  
"50dfe4ad9b1d04bb8065625460b3e8ca27f08cf824cb2dfd21cf70e843ff02e4", "locator": "lines35\\u201346,69\\u201378",  
"observation": "engineered cosmo_rank before split"}
```

E-LCL-013 - Random-feature baseline targets stellar mass and named input lacks target

[Certain] model5.py targets logmstar, not SFR. The discovered SDSSDR18_200000.csv has 18 columns and lacks logmstar, so the named input cannot run the provided target expression as-is. The source samples 5 features from all columns without excluding the target. Reconstructing the declared seed 42 feature sampling on the as-found header gives objid,ra,run,r,u; target inclusion is not confirmed for this exact input.

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\model5.py", "sha256":  
"7d1d9e3d57e26e37ec4e1a9f9578b9ff5d4291f9e68a66eb327a50f638095d44", "locator":  
"lines12,16\\u201322,26,32,37", "observation": "target logmstar; no target exclusion"}  
  
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\SDSSDR18_200000.csv", "sha256":  
"9dca2008683fde4b0f7a90af5de85ee846a30a47f6f6fae3968c83dea06fa5d2", "locator": "CSVheader line1",  
"observation": "18columns, no logmstar"}
```

E-LCL-014 - Local CSV registries reuse canonical IDs with conflicting meanings

[Certain] Repository registry CSVs use CLAIM-ACSC-001 etc rather than PDF CORE-001 etc. More seriously, experiment CSV maps EXP-CTRL-A01 to provenance control, EXP-CTRL-A02 to leakage control, contrary to attached PDF random-feature/permutation definitions. Dataset entries remain planned with deferred provenance. These namespaces cannot be silently merged by experiment-string identity.

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\registry\\claim_evidence_v0.2.csv", "sha256":  
"ef50ff75fe995633934e7a12a635f76d16333ed7d260ddd1fee7930bb9d433f", "locator": "CSVheader  
andlines2\\u20137", "observation": ""}  
  
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\registry\\experiment_registry_v0.2.csv", "sha256":  
"b2a8c69710fedfa708d763eebf150cf8b35fe49da8ef8392e9ee0a4625161399", "locator": "CSVlines4\\u20136",  
"observation": ""}  
  
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\registry\\claim_experiment_crosswalk_v0.2.csv",  
"sha256": "d8b5bc55187b1ba0f2e6d1e2f0d2a1571c2e92a8e93570de25191892290e34ec", "locator":  
"CSVlines4\\u20136", "observation": ""}  
  
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\S.T.A.R.-Program\\registry\\dataset_registry_v0.1.csv", "sha256":  
"b11f2cf25b903ff1ad32c7d35a6353b3f327d31133f8c816363f26865f78150b", "locator": "CSVlines2\\u20139",  
"observation": ""}
```

E-LCL-015 - Partial upstream download metadata survives for observational sources

[Certain] Download-zone sidecars identify MaNGA HI DR17 v2_0_1 FITS and a SkyServer/CasJobs SDSS CSV export. They provide partial upstream attribution; they do not include the full SDSS SQL, release/query selection, acquisition hash, or complete raw-to-model lineage. The ecdata README identifies the Cremona database and a Zenodo release DOI, but the selected derived arithmetic manifests do not bind to a specific commit/release.

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\mangaHlall.fits\\uf03aZone.Identifier", "sha256":  
"de193697f18a34335d7a0c99b8d4fbb3ab730cd3ca76e83a667e2aa3aeb8dfc0", "locator": "ZoneTransfer ReferrerUrl  
and HostUrl", "observation": "https://data.sdss.org/sas/dr17/manga/HI/v2_0_1/mangaHlall.fits"}
```

```
{"path": "C:\\Users\\pmqr7\\OneDrive\\Documents\\STAR\\SDSSDR18_200000.csv\\uf03aZone.Identifier", "sha256":  
"91ff0b446804f8dfe2aa0e2066df08629b2a4ac12a7f9f6b28825679af038b84", "locator": "ZoneTransfer HostUrl",  
"observation": "https://skyservice.pha.jhu.edu/CasJobsOutput/CSV/MyTable_pmqr771_0.csv"}
```

```
{"path": "\\wsl.localhost\\Ubuntu\\home\\kepler\\ecdata\\README.md", "sha256":  
"619466f9a1542c3bfeaa68b592feb903edd205c652cac59d7f21d70444203bb5", "locator": "lines4\\u201312",  
"observation": "Cremona Database; DOI10.5281/zenodo.161341"}
```

E-LCL-016 - STAR map visualization uses illustrative synthetic data and preselected Hubble calibration

[Certain] Recovered S.T.A.R.M.A.P. catalog explicitly labels its data illustrative, combines 12 hardcoded example curves with 148 synthetically generated rows, and seeds generation with 0x53544152. The H_eff implementation states calibration so high-z recovers Planck and low-z approaches SH0ES, with hardcoded correction coefficients. Geometry angles are hash-derived from curve labels; cluster summaries are hardcoded. These visualizations do not supply independent observational or arithmetic validation.

```
{"path": "/home/kepler/S.T.A.R.M.A.P./src/lib/star/catalog.ts", "readable_audit_copy": "C:/Users/pmqr7/.codex/.chatgpt-p  
rojects/g-p-6aa87346c5e88191979368ee9ee5ce5b/audit_work/recovered_sources/recovered_00088.ts", "sha256":  
"f585130c982be4789c4bc701119b1a1a5fdcb72bd475700b01d48c8c921fbb29", "locator":  
"lines27\\u201340,43\\u2013387,91\\u2013399", "observation": "Recovered by native WSL read; original source preserved"}
```

```
{"path": "/home/kepler/S.T.A.R.M.A.P./src/lib/star/physics.ts", "readable_audit_copy": "C:/Users/pmqr7/.codex/.chatgpt-p  
rojects/g-p-6aa87346c5e88191979368ee9ee5ce5b/audit_work/recovered_sources/recovered_00085.ts", "sha256":  
"a9125fdcf3570012c8f1a3467562fecf31132fecf301fd3decdf300e59bfe234", "locator": "lines27\\u201342,117\\u2013126",  
"observation": "Recovered by native WSL read; original source preserved"}
```

```
{"path": "/home/kepler/S.T.A.R.M.A.P./src/routes/hubble.tsx", "readable_audit_copy": "C:/Users/pmqr7/.codex/.chatgpt-p  
rojects/g-p-6aa87346c5e88191979368ee9ee5ce5b/audit_work/recovered_sources/recovered_00063.tsx", "sha256":  
"0ac7b166116e1d3ae57fcc57a1e9435a4e5c7b3d26d70dc34349601c8b24b24d", "locator":  
"lines13\\u201343,125\\u2013149", "observation": "Recovered by native WSL read; original source preserved"}
```

```
{"path": "/home/kepler/S.T.A.R.M.A.P./README.md", "readable_audit_copy": "C:/Users/pmqr7/.codex/.chatgpt-projects/  
g-p-6aa87346c5e88191979368ee9ee5ce5b/audit_work/recovered_sources/recovered_00005.md", "sha256":  
"e51bbff49db1e33900f05f5082985a576f18a0de8580e733a78e75680dacc7e3", "locator": "Vision and Implementation  
Status sections", "observation": "Recovered by native WSL read; original source preserved"}
```

E-DRV-001 - K convergence and unstable small-sample values are historical theoretical/simulation reports

[Certain] The Cosmological L-function source describes a log-normal expectation argument and reports $K=0.01176$ for $N=9$ and $K_{\text{sub}}=0.00322$ for $N=5$. These are retained as theoretical/historical and negative/diagnostic provenance; this audit did not rerun the simulations or establish a BSD/cosmological physical correspondence.

```
{ "drive_id": "1lzJvyyI_WSAGDe--IdU8LhOupC3fU8qLG6fzcFbSJSY", "title":  
"Cosmological_L-function+Galaxy_Mass_Distribution", "url":  
"https://docs.google.com/document/d/1lzJvyyI_WSAGDe--IdU8LhOupC3fU8qLG6fzcFbSJSY/edit", "locator": "lines  
86-91, 116-122", "snapshot_text_sha256":  
"e056655477defaaf2eaa4cec28f4a3d564ec408fd6f333bb1cae353d995383f3", "digest_type": "SHA256 of  
connector-readable UTF-8 text snapshot, not canonical provider binary or revision digest", "snapshot_file":  
"1lzJvyyI_WSAGDe--IdU8LhOupC3fU8qLG6fzcFbSJSY.txt" }
```

E-DRV-002 - Drive SFR source contains preprocessing before the split in inspected script variants

[Certain] The SFR Prediction Model Refinement snapshot contains an imputer fit_transform at line 140 before train_test_split at line 161, and other whole-data feature construction variants. This is a static validation-design concern for those variants; execution of that exact variant and the magnitude of any performance effect were not established.

```
{ "drive_id": "1bzyY4bg5_Vi3RucO7NllkPtYpl8GptmqIRtP_WF_qaw", "title": "SFR_Prediction_Model_Refinement", "url":  
"https://docs.google.com/document/d/1bzyY4bg5_Vi3RucO7NllkPtYpl8GptmqIRtP_WF_qaw/edit", "locator": "lines  
117-166, 413-424", "snapshot_text_sha256":  
"d2d83ffc304678df0813d76491dc8442789f0c831f47f6d120029cdcf41ac2a3", "digest_type": "SHA256 of  
connector-readable UTF-8 text snapshot, not canonical provider binary or revision digest", "snapshot_file":  
"1bzyY4bg5_Vi3RucO7NllkPtYpl8GptmqIRtP_WF_qaw.txt" }
```

E-DRV-003 - Five-fold validation document does not itself establish independent validation

[Certain] The 5 Fold Validation document contains preprocessing/derived-statistic code and prose describing cross-validation as robustness. A readable document or a stated five-fold procedure is not a linked independent dataset execution; the inspected material does not satisfy the charter full data-code-parameters-null-results chain by itself.

```
{ "drive_id": "1qSkMYPbRGytMD4qEB-llHcBt_V3cXb4nAQs7JtAo96U", "title": "5 Fold Validation", "url":  
"https://docs.google.com/document/d/1qSkMYPbRGytMD4qEB-llHcBt_V3cXb4nAQs7JtAo96U/edit", "locator": "lines  
1-48", "snapshot_text_sha256": "5056614627b541bc35432235a1636b9d87afd8e4c9642c3f335c56d62013ee36",  
"digest_type": "SHA256 of connector-readable UTF-8 text snapshot, not canonical provider binary or revision digest",  
"snapshot_file": "1qSkMYPbRGytMD4qEB-llHcBt_V3cXb4nAQs7JtAo96U.txt" }
```

E-DRV-004 - Drive raw data log mixes code proposals and conversational interpretations

[Certain] The raw data log includes research interpretations, script variants and conversational requests to run future steps. These establish documentary provenance, not authenticated executions. Source directions were treated as source text and were not followed.

```
{ "drive_id": "1ZVWuuwAS8zRrEsLIYNNSFNuZRGjUWdHbl4WSxpFVreo", "title": "raw data log", "url":  
"https://docs.google.com/document/d/1ZVWuuwAS8zRrEsLIYNNSFNuZRGjUWdHbl4WSxpFVreo/edit", "locator": "lines  
11-15, 48-187, 2438-2472", "snapshot_text_sha256":  
"ccfc95874403c889504493591f95f618ba69ec3421e433c6eb2fbd9fee19c27d", "digest_type": "SHA256 of  
connector-readable UTF-8 text snapshot, not canonical provider binary or revision digest", "snapshot_file":  
"1ZVWuuwAS8zRrEsLIYNNSFNuZRGjUWdHbl4WSxpFVreo.txt" }
```

Retained v0.2 baseline follows

The following pages reproduce the complete supplied reference PDF. Its claims, experiment methods and historical interpretations remain traceable. Consult the current audit supplement for corrections and evidence maturity.

S.T.A.R.

Claim → Evidence → Experiment Crosswalk

Purpose

This crosswalk connects the v0.2 Claim/Evidence Registry to the existing S.T.A.R. research and development archive.

It answers four questions for every active claim:

1. **What existing R&D evidence can be used now?**
2. **What evidence requires reconstruction or validation before use?**
3. **What experiment formally tests the claim?**
4. **Is the next step reconstruction, controlled re-analysis, or an entirely new experiment?**

The crosswalk deliberately distinguishes:

```
[  
  \boxed{\text{Existing Evidence}}  
  \neq  
  \boxed{\text{Validated Evidence}}  
]
```

Existing R&D can therefore be scientifically useful while remaining exploratory, historical, or reconstruction-required.

1. Evidence-State Legend

Code	Evidence state	Meaning
U	Usable R&D Evidence	Existing artifact can enter the evidence chain as historical, theoretical, computational, or pilot evidence

R	Reconstruction Required	Existing result is potentially valuable but methodology/data/code must be reconstructed before controlled use
N	New Experiment Required	Existing R&D does not adequately test the claim
C	Control	Existing material can provide a baseline/null/control
D	Derivation Audit	Mathematical derivation exists but requires formal verification
H	Historical Only	Preserved for provenance; not evidence for current claim
X	Negative/Diagnostic	Existing result identifies instability, limitation, or contradictory behavior

These states may coexist.

For example:

[
 $\boxed{U+R}$
]

means useful historical evidence exists, but reconstruction is required before promotion to controlled evidence.

2. Overall Crosswalk

Claim	Existing R&D Evidence	State	Formal Experiment	Next Action
CORE-001	Multiple mapping/topology/model documents	U + R	EXP-CORE-A 01	Reconstruction + new controlled test
RANK-001	Rank/mapping theory + historical mapping work	U + N	EXP-RANK-A 01	New experiment

RANK-002	Persistent-homology methodology + mapping framework	U + N	EXP-RANK-A02	New experiment
RANK-003	Historical rank/environment hypothesis	H	EXP-RANK-A03	New experiment
MAP-001	Alternative-mapping document	U	EXP-MAP-A01	Controlled comparison
MAP-002	Historical ($\backslash\Phi$) implementations	U + R	EXP-MAP-A01	Reconstruction
MAP-003	Historical (z=200r) work	U + R	EXP-MAP-A02	Parameter sweep
MAP-004	Criteria for Alternative Mappings	U	EXP-MAP-A01	Formalize test
MAP-005	MCJ/PTD/FT alternatives	U	EXP-MAP-A01	New controlled comparison
ARITH-001	Fibonacci/Lucas implementations	U + H	EXP-MAP-A03	New control experiment
TOPO-001	TDA code/methodology + historical distance results	U + R	EXP-TOPO-A01	Reconstruction
TOPO-002	Void/topology framework	H	EXP-TOPO-A02	New experiment
TOPO-003	Connected-component methodology	U	EXP-TOPO-A03	New experiment
MATH-K001	Validation of BSDCT.pdf + simulations	U + R	EXP-K001	Mathematical reconstruction
MATH-K002	($K\backslash\text{approx}0.998108$) simulations	U	EXP-K001	Reproduce
MATH-K003	Conflicting historical K values	U + X	EXP-K002	Stability study
COSMO-L001	L-cosmo(s).py and related scripts	U + R	EXP-L001	Leakage-controlled reconstruction

COSMO-L002	Historical SFR/L-cosmo models	H + R	EXP-SFR-A01	New controlled test
COSMO-R001	<code>cosmo_rank</code> implementations	U + H	CTRL-COSMO-001	Rename/reconstruct
SFR-A001	Historical RF/GB/SHAP models	U + R	EXP-SFR-A01	New controlled test
SFR-A002	Historical feature/model work	H	EXP-SFR-A02	New experiment
SFR-E001	Historical outlier results	U	EXP-SFR-A03	Reproduce
SFR-A003	Existing outlier analysis	U	EXP-SFR-A03	Pre-register
CTRL-RF001	<code>model15.py</code>	U + C	EXP-CTRL-A01	Reproduce
CTRL-A001	No adequate completed permutation control identified	N	EXP-CTRL-A02	New experiment
CTRL-A002	Leakage concerns identified in scripts	X	EXP-CTRL-A03	Mandatory reconstruction
DATA-M001	<code>merged_sdss_hi.csv</code> + KDTREE workflow	U	EXP-DATA-A01	Reproduce matching
DATA-M002	Extensive cross-survey datasets	U	DATA-PROV-001	Formalize provenance
RTCH-M001	RTCH manuscript/action	U + D	EXP-RTCH-B01	Derivation audit
RTCH-M002	Closed-pullback construction	U + D	EXP-RTCH-B01	Formal proof check
RTCH-H001	Candidate coupling equations	U	EXP-RTCH-B02	New physical test
RTCH-H002	Trace-coupling derivation	U + D	EXP-RTCH-B03	Derivation + test
RTCH-H003	Point-particle action/equation	U + D	EXP-RTCH-B02	Constraint analysis

RTCH-H004	Unified-matter construction	U	EXP-RTCH-B01	Formal consistency test
FIELD-001	Arithmetic-topography formulation	U	EXP-FIELD-B01	Formalize
FIELD-002	No controlled physical evidence	N	EXP-FIELD-B02	New experiment
FIELD-003	Symbolic-field proposal	H	EXP-FIELD-B03	New theoretical construction
ECC-001	ECC manuscript	U + D	EXP-ECC-B01	Formal derivation
ECC-002	Closed/non-exact form construction	U + D	EXP-ECC-B01	Mathematical audit
ECC-003	Proposed entropy-energy architecture	H	EXP-ECC-B02	New theory/limit test
ENERGY-001	Conceptual mass-energy proposals	H	EXP-ENERGY-B01	New derivation
COSMO-E001	Redshift/topology framework	H	EXP-COSMO-B01	New longitudinal experiment
COSMO-E002	No direct evidence	N	EXP-COSMO-B02	Downstream only

3. CORE-001 — Arithmetic–Cosmic Structural Correspondence

Claim

Arithmetic invariants may map into geometric/topological structures producing statistically nontrivial correspondence with independent astronomical data.

Existing Evidence

The archive contains:

- [Criteria_for_Alternative_Mappings_Φ.pdf](#)
- [The_Arithmetic-Cosmic_Structure_Conjecture_\(ACSC\)_Monograph.pdf](#)
- [_Arithmetic_Invariants_and_Cosmological_Geometry_in_Cartography_.pdf](#)
- historical mapping scripts;
- TDA/persistent-homology development;
- astronomical datasets;
- historical correspondence calculations.

Evidence State

U + R

The existence of the mathematical construction and historical computational work is usable as provenance.

The historical numerical correspondence results require reconstruction.

Experiment

EXP-CORE-A01 — Arithmetic/Cosmic Independent Correspondence

Pipeline:

```
[
E
\rightarrow I(E)
\rightarrow \Phi(E)
\rightarrow \mathcal{H}_{\rm arith}
\rightarrow D_{\rm arith}
\rightarrow D_{\rm cosmic}.
]
```

Required Controls

- arithmetic permutation;
- alternative invariant;
- alternative mapping;
- spatial randomization;
- astrophysical baseline.

Decision

Reconstruction + genuinely new controlled experiment.

4. RANK-001 — Rank/Environment Correspondence

Existing Evidence

Historical S.T.A.R. documents explicitly propose rank/environment relationships.

Historical mapping work provides candidate rank-based coordinates.

Evidence State

U as hypothesis provenance.

There is not yet sufficient controlled evidence establishing the relationship.

Experiment

EXP-RANK-A01 — Rank/Environment Association

Test:

[
P($r_{E \mid \text{environment}}$)
]

against invariant and permutation controls.

Primary Variables

- elliptic-curve rank;
- environment class;
- density;
- morphology;
- cluster/filament/void classification.

Status

N — New experiment required.

5. RANK-002 — Rank/Topological Complexity

Existing Evidence

The archive contains:

- persistent-homology methodology;
- arithmetic mapping constructions;
- cosmic-structure topology discussions;
- TDA computational development.

Evidence State

U + N

The methodological foundation is usable, but the core rank/topology experiment is not established by the existing archive.

Experiment

EXP-RANK-A02 — Rank-Weighted Persistent Homology

Compare:

[
 $D_{\text{arith}}^{(r)}$
]

against observed cosmic topology.

Use:

[
 $\beta_0, \beta_1, \beta_2$
]

plus persistence distances.

Status

New controlled experiment required.

6. RANK-003 — Rank-Stratified Cosmic Structure

Historical rank classes such as:

[
r=0,1,2,3
]

were associated with increasingly complex cosmic environments.

Evidence State

H — historical hypothesis.

Experiment

EXP-RANK-A03 — Rank-Stratified Environment Test

Pre-register rank strata before examining cosmic outcomes.

Status

New experiment.

7. MAP-001 — Arithmetic Projection

Existing Evidence

Strongest current documentary support is:

[Criteria_for_Alternative_Mappings_Φ.pdf](#).

It defines:

[
\\Phi(E)=
(\\log|\\Delta|,\\log N,\\alpha r).
]

Evidence State

U

The mapping itself is an existing mathematical/model construction.

Experiment

EXP-MAP-A01 — Mapping Family Comparison

Test $(\backslash\Phi)$ against alternative mappings.

Status

Existing construction usable; controlled comparison required.

8. MAP-002 — Extended Projection

Historical work uses:

$(\backslash\phi_E, \backslash\theta_E, z_E, s_E)$.
]

Evidence

Existing scripts and theoretical documents.

State

U + R

Experiment

EXP-MAP-A01

Reconstruct the exact historical transformation and then test it against alternatives.

Status

Reconstruction required.

9. MAP-003 — ($z=200r$)

Historical formulation:

[
 $z=200r$.
]

Existing Evidence

Multiple historical mapping documents/scripts.

State

U + R

The existence of the mapping is established.

Its physical justification is not.

Experiment

EXP-MAP-A02 — Rank Scaling Sensitivity

Test:

[
 $z=\lambda_r r$
]

over a pre-registered range of (λ_r).

Compare (200) against alternative values without selecting the value based on test-set performance.

Status

New controlled parameter experiment.

10. MAP-004 / MAP-005 — Alternative Mappings

Existing Evidence

[Criteria_for_Alternative_Mappings_Φ.pdf](#) explicitly supports:

- no strict bijection requirement;
- equivalence classes;
- MCJ;
- PTD;
- FT;
- alternative mapping families.

State

U

This is one of the strongest areas of existing documentary support.

Experiment

EXP-MAP-A01

Required Outcome

Determine whether correspondence is:

1. mapping-specific;
2. invariant-specific;
3. generic to flexible transformations.

Status

Ready for formal experimentalization.

11. ARITH-001 — Fibonacci/Lucas Generator Family

Existing Evidence

Historical mapping implementations use Fibonacci/Lucas structures.

State

U + H

The computational construction exists.

Physical significance is unestablished.

Experiment

EXP-MAP-A03 — Arithmetic Generator Control

Compare:

[
 \text{Fibonacci/Lucas}
]

against:

[
 \text{primes},
 \quad
 \text{random recurrences},
 \quad
 \text{matched integer sequences}.
]

Status

New experiment required.

12. TOPO-001 — Persistent-Homology Correspondence

Existing Evidence

R&D includes TDA methodology and historical Wasserstein-distance claims.

Historical threshold:

[
 $W_2 < 10^{-2}$.
]

State

U + R

The methodology is usable.

The historical threshold is not accepted as a universal criterion.

Experiment

EXP-TOPO-A01 — Null-Calibrated Topological Correspondence

Build the empirical null distribution of Wasserstein/bottleneck distances.

Status

Reconstruction + controlled experiment.

13. TOPO-002 — Void Structure

Existing Evidence

Theoretical framework connects topology to void structure.

State

H / H

There is insufficient controlled empirical evidence.

Experiment

EXP-TOPO-A02 — Arithmetic/Voids Test

Compare arithmetic-derived (β_2) structure with independently identified voids.

Status

New experiment.

14. TOPO-003 — Connected Components

Existing Evidence

Persistent homology provides (β_0) .

State

U methodological evidence.

Experiment

EXP-TOPO-A03 — Connected-Component Test

Compare:

[
 β_0^{arith}
]

with observed component/cluster structure.

Status

New controlled experiment.

15. MATH-K001 — $(K_{\text{theory}} \rightarrow 1)$

Existing Evidence

Validation of BSDCT.pdf contains:

$$\frac{\sum_i(M_i/M_0)}{\sum_i(M_0/M_i)}$$

and the log-normal argument.

Simulations used:

[
 $N=500$,
 $\mu=9.7843$,
 $\sigma=0.3645$.
]

State

U + R

The mathematical idea is usable.

The exact proof and simulation pipeline should be reconstructed.

Experiment

EXP-K001 — Log-Normal Reciprocal-Normalization Test

Status

Reconstruct and independently reproduce.

16. MATH-K002 — Simulation ($K \approx 0.998108$)

Existing Evidence

Historical simulations report:

[
 $K \approx 0.998108$.
]

State

U as historical simulation evidence.

Experiment

EXP-K001

Required Test

Reproduce the simulation independently.

Status

Reproduction required.

17. MATH-K003 — K Instability

Historical R&D also reports:

[
K\approx0.01176
]

for one analysis and approximately:

[
K\approx0.00322
]

for a subsample.

State

U + X

This is important negative/diagnostic evidence.

Experiment

EXP-K002 — K Sensitivity and Stability

Investigate:

- sample size;
- distribution;
- filtering;
- redshift;
- mass definition;
- normalization;
- outliers.

Status

High-value reconstruction experiment.

18. COSMO-L001 — ($L_{\rm cosmo}(s)$)

Existing Evidence

`L-cosmo(s).py` and related scripts implement:

```
\sum_i  
\left(\frac{M_0}{M_i}\right)^s.  
]
```

They also construct:

- mass-dependent quantities;
- redshift bins;
- morphology-derived variables;
- SFR targets;
- RF/GB models;
- SHAP analyses.

State

U + R

This is a substantial existing development line.

Critical Problem

Some derived quantities appear to be constructed before train/test splitting.

Therefore historical model performance cannot automatically be promoted to controlled evidence.

Experiment

EXP-L001 — Leakage-Controlled ($L_{\rm cosmo}$) Validation

Status

Reconstruct first.

19. COSMO-L002 — ($L_{\rm cosmo}$) and SFR

Existing Evidence

Historical SFR scripts use ($L_{\rm cosmo}$)-related variables.

State

H + R

Experiment

EXP-SFR-A01

Comparison

[
 $X_{\rm astro}$
]

versus

[
 $X_{\rm astro} + X_{L_{\rm cosmo}}$.
]

Status

New controlled experiment after reconstruction.

20. COSMO-R001 — **cosmo_rank**

Historical scripts construct **cosmo_rank** from morphology/redshift variables.

Evidence State

U + H

Important Registry Action

Rename this quantity so that it cannot be confused with elliptic-curve rank.

Recommended:

```
[  
R_{\rm cosmo}^{\rm feat}.  
]
```

Experiment

CTRL-COSMO-001 — Composite-Feature Ablation

Test whether the composite feature adds information beyond its constituent variables.

Status

Reconstruction/control.

21. SFR-A001 — Arithmetic Incremental Prediction

Existing Evidence

Substantial SFR ML development exists:

- RF;
- GB;
- SHAP;
- Optuna;
- cross-validation;
- feature interactions;
- Pipe3D/MaNGA/SDSS datasets.

Historical performance includes approximately:

[
 $R^2_{\rm RF}=0.8455$
]

and

[
 $R^2_{\rm GB}=0.8476$.
]

State

U + R

These establish an astrophysical ML development baseline.

They do not establish arithmetic predictive power.

Experiment

EXP-SFR-A01

Compare:

[
 $X_{\rm astro}$
]

against:

[
 $X_{\rm astro}+X_{\rm arith}$.
]

Primary Endpoint

[
 $\Delta R^2_{\rm OOS}$.
]

Status

New controlled experiment required.

22. SFR-A002 — Arithmetic-Only SFR

Existing Evidence

No adequate controlled arithmetic-only result has been identified.

State

N

Experiment

EXP-SFR-A02

[
 $Y=f(X_{\rm arith})$.
]

Compare against null and astrophysical baseline.

Status

Entirely new experiment.

23. SFR-E001 — Extreme-SFR Behavior

Historical results distinguish:

- full sample;
- outlier population;
- non-outlier population.

State

U

This is directly usable as historical empirical/model evidence.

Experiment

EXP-SFR-A03

Status

Reproduction required.

24. CTRL-RF001 — Random Feature Baseline

Existing Evidence

`model15.py` uses randomly selected features.

State

U + C

Experiment

EXP-CTRL-A01

Reproduce with locked random seeds and identical evaluation design.

Purpose

Determine how much apparent performance can arise from model flexibility alone.

Status

Ready for formal control reconstruction.

25. CTRL-A001 — Arithmetic Permutation

Existing Evidence

No adequate completed permutation experiment was identified in the audit.

State

N

Experiment

EXP-CTRL-A02

Permute arithmetic assignments while preserving:

- marginal distributions;
- sample size;
- missingness where appropriate.

Status

Entirely new experiment.

26. CTRL-A002 — Leakage-Controlled Reconstruction

Existing Evidence

The audit identified possible leakage in:

- normalization;
- redshift binning;
- ($L_{\rm cosmo}$);
- `cosmo_rank`;

- interaction construction.

State

X — diagnostic evidence.

Experiment

EXP-CTRL-A03 — Leakage Audit

Required Procedure

Rebuild preprocessing within folds.

Status

Mandatory before promotion of historical ML results.

27. DATA-M001 — Two-Arcsecond Matching

Existing Evidence

`merged_sdss_hi.csv` was produced through coordinate matching using KDTree and approximately:

```
[  
  2"  
]
```

matching tolerance.

`arcsec_separation` was retained.

State

U

Experiment

EXP-DATA-A01 — Cross-Survey Matching Reproduction

Test matching sensitivity to:

```
[  
1",2",3",5"  
]
```

and inspect false-match/duplicate behavior.

Status

Reproduction/validation.

28. DATA-M002 — Multi-Survey Infrastructure

Existing Evidence

The archive contains substantial:

- SDSS;
- MaNGA;
- Pipe3D;
- HI;
- Galaxy Zoo;
- stellar-mass;
- environment;
- 2MASS;
- UKIDSS;
- VizieR

data resources.

State

U

Experiment

DATA-PROV-001 — Dataset Lineage Audit

Required Output

```
[  
D_{\rm raw}  
\rightarrow  
D_{\rm matched}  
\rightarrow  
D_{\rm analysis}  
\rightarrow  
D_{\rm model}  
\rightarrow  
D_{\rm test}.  
]
```

Status

Ready for formal registry construction.

29. RTCH-M001 — Variational RTCH Construction

Existing Evidence

The RTCH manuscript contains:

- proposed action;
- fields;
- matter coupling;
- field equations;
- point-particle dynamics;
- perfect-fluid dynamics;
- topological sector;
- observational tests;
- falsification framework.

State

U + D

The proposed mathematical architecture exists.

Experiment

EXP-RTCH-B01 — RTCH Mathematical Consistency Audit

Verify:

- variation with respect to $(g_{\mu\nu})$;
- variation with respect to (ϕ) ;
- variation with respect to (Y) ;
- B-field equations;
- boundary terms;
- conservation laws;
- dimensions;
- limiting cases.

Status

Derivation audit required.

30. RTCH-M002 — Closed Pullback

Existing Evidence

RTCH defines:

$$\begin{bmatrix} F_Q = \Phi^* \omega \end{bmatrix}$$

with

$$\begin{bmatrix} d_{\mathcal{T}} \omega = 0. \end{bmatrix}$$

Therefore:

$$\begin{bmatrix} dF_Q = 0. \end{bmatrix}$$

State

U + D

Experiment

EXP-RTCH-B01

Status

Formal proof/derivation audit.

This is among the claims closest to mathematical verification because it concerns the stated differential-geometric construction rather than an empirical cosmological assertion.

31. RTCH-H001 — Topological Coupling

Existing Evidence

Candidate coupling exists in the RTCH framework.

State

U as theoretical construction.

No observational validation currently established.

Experiment

EXP-RTCH-B02 — Fifth-Force/Topological Coupling Constraints

Status

New physical experiment/constraint analysis.

32. RTCH-H002 — Radiation Trace

Existing Evidence

RTCH derives:

[
 $\bar{T} = -\epsilon + 3p.$
]

For

[
 $p = \epsilon/3,$
]

therefore:

[
 $\bar{T} = 0.$
]

State

U + D

Experiment

EXP-RTCH-B03

Verify the result from the complete action and examine its physical implications.

Status

Derivation audit + physical consistency test.

33. RTCH-H003 — Fifth-Force Acceleration

Existing RTCH formulation:

[
 $S_p = -mc \int A, ds$
]

leading to:

[
 $-c^2 h^{\{\mu \nu\}} \nabla_\nu \ln A.$
]

State

U + D

Experiment

EXP-RTCH-B02

Compare predicted deviations with established gravitational/Equivalence Principle constraints.

Status

New physical constraint experiment.

34. FIELD-001 — Arithmetic Topography

Existing Evidence

RTCH/ACSC architecture proposes:

$f(l_1, \ldots, l_n)$
]

and

$\nabla h_{\rm arith}$.
]

State

U

As a mathematical candidate construction.

Experiment

EXP-FIELD-B01 — Arithmetic Topography Construction

First determine whether the field can be defined consistently and reproducibly.

Status

Formalization required.

35. FIELD-002 — Arithmetic Field as Gravity

Existing Evidence

Conceptual proposals exist.

State

N

No controlled physical evidence identified.

Experiment

EXP-FIELD-B02 — Gravitational Field Comparison

Compare arithmetic field predictions against:

- density;
- lensing;
- dynamics;
- peculiar velocities.

Status

Entirely new physical experiment.

36. FIELD-003 — Arithmetic Tensor

Existing Evidence

Historical ACSC material proposes symbolic field equations.

State

H

Experiment

EXP-FIELD-B03

Construct a mathematically consistent tensor theory before attempting physical comparison.

Status

New theoretical development.

37. ECC-001 — Entropy/Cohomology Architecture

Existing Evidence

[The_Entropy_Cohomology_Conjecture_\(ECC\).pdf](#).

State

U + D

The conceptual and mathematical architecture exists.

Experiment

EXP-ECC-B01

Formal derivation and cohomological consistency audit.

Status

Reconstruction/derivation.

38. ECC-002 — Closed Non-Exact Requirement

Existing Evidence

The revised ECC framework uses:

[
d\omega=0,
\quad
[\omega]\neq0.
]

State

U + D

Experiment

EXP-ECC-B01

Status

Mathematical verification.

39. ECC-003 — Entropy-Energy Relation

Existing Evidence

Conceptual proposals exist.

State

H / H

Experiment

EXP-ECC-B02

Required:

[
 \text{derivation}
 \rightarrow
 \text{dimensional analysis}
 \rightarrow
 \text{known limit}
 \rightarrow
 \text{conservation}
 \rightarrow
 \text{novel prediction}.
]

Status

Entirely new theoretical/experimental program.

40. ENERGY-001 — Generalized Mass-Energy Relation

Existing Evidence

Historical theoretical proposals.

State

H

Experiment

EXP-ENERGY-B01

Require:

[
 $E_{\rm STAR}$
 $\rightarrow$
 mc^2
]

in the appropriate known limit.

Status

New theoretical construction.

41. COSMO-E001 — Redshift Evolution

Existing Evidence

The archive contains:

- redshift-dependent ($L_{\rm cosmo}$);
- redshift bins;
- cosmological mapping ideas;
- persistent-homology framework.

State

U + H

Experiment

EXP-COSMO-B01 — Redshift-Resolved Arithmetic Topology

Compare topology across redshift bins.

Status

New controlled longitudinal experiment.

42. COSMO-E002 — Hubble–Planck Tension

Existing Evidence

Theoretical motivation exists.

State

N

There is no adequate direct evidence connecting the current S.T.A.R. framework to the tension.

Experiment

EXP-COSMO-B02

Only activate after the lower-level arithmetic/topological/dynamical claims have demonstrated sufficient empirical support.

Status

Downstream / deferred.

43. Evidence Maturity Matrix

The audit allows the current R&D to be grouped into four broad categories.

Category A — Already Usable as R&D Evidence

These artifacts can immediately populate the evidence registry, provided they are labeled correctly:

- alternative mapping construction;
- historical (Φ) definitions;
- RTCH mathematical architecture;
- ECC mathematical architecture;
- $(L_{\rm cosmo})$ implementation;
- SFR model baselines;
- random-feature model;
- historical extreme-SFR results;
- cross-survey matching workflow;
- existing datasets;
- historical contradictory (K) results.

These establish **research provenance and candidate constructions**, not final scientific conclusions.

44. Category B — Valuable but Requires Reconstruction

The highest-priority reconstruction targets are:

B1 — Historical (K)

Reconstruct:

```
[  
K_{\rm theory}  
]
```

from the original simulation assumptions.

B2 — ($L_{\rm cosmo}$)

Rebuild the feature-generation process fold-by-fold.

B3 — Historical SFR models

Reproduce historical RF/GB results with locked datasets and leakage control.

B4 — Historical topological distances

Reconstruct the exact point clouds, normalization, filtration, and persistence calculations.

B5 — Historical ($z=200r$)

Reproduce the mapping independently and test sensitivity to (λ_r).

B6 — Historical (Φ)

Reconstruct exact arithmetic-to-coordinate transformations.

B7 — **cosmo_rank**

Separate morphology/redshift composite information from true arithmetic rank.

45. Category C — Requires Entirely New Experiments

The following claims are not adequately tested by the existing R&D:

1. rank $\rightarrow$ cosmic environment;
2. rank $\rightarrow$ topology;
3. arithmetic-only SFR prediction;
4. arithmetic incremental SFR information;
5. arithmetic permutation null;
6. Fibonacci/Lucas versus control sequences;
7. arithmetic field $\rightarrow$ gravity;
8. RTCH $\rightarrow$ fifth-force observations;
9. entropy/cohomology $\rightarrow$ energy;
10. arithmetic topology $\rightarrow$ cosmic evolution.

These should not be “validated” by reinterpreting historical results.

They require registered experiments.

46. Highest-Priority Reconstruction Queue

The recommended reconstruction order is:

R-001

Leakage Audit

All historical ML results.

R-002

SFR Baseline Reconstruction

Reproduce conventional RF/GB performance.

R-003

($L_{\rm cosmo}$) Reconstruction

Rebuild without leakage.

R-004

Arithmetic Feature Reconstruction

Separate genuine arithmetic variables from engineered morphology/redshift variables.

R-005

K Reconstruction

Reproduce mathematical/statistical result.

R-006

Mapping Reconstruction

Reproduce (Φ), ($z=200r$), MCJ/PTD/FT.

R-007

Topological Reconstruction

Reproduce historical persistence calculations.

47. Highest-Priority New Experiments

After the reconstruction queue:

A-001

EXP-CTRL-A02 — Arithmetic Permutation Control

This is essential because it establishes whether apparent arithmetic signal survives destruction of object-level association.

A-002

EXP-SFR-A01 — Incremental Arithmetic SFR

This directly tests the project's SFR objective.

A-003

EXP-RANK-A01 — Rank/Environment

This directly tests the primary arithmetic/cosmic-structure hypothesis.

A-004

EXP-RANK-A02 — Rank/Topology

This moves the test from ordinary statistics into the intended TDA framework.

A-005

EXP-MAP-A01 — Mapping Family Comparison

This tests whether the result depends on a particular hand-selected mapping.

48. Critical Dependency Graph

The experiments should not be treated as independent.

The current dependency structure is:

```
[  
  \boxed{  
    \text{Data Provenance}  
  }  
]
```

```
[  
  \downarrow  
]
```

```
[  
  \boxed{  
    \text{Leakage / Matching Validation}  
  }  
]
```

}
]

[
\downarrow
]

[
\boxed{
\text{Astrophysical Baselines}
}
]

[
\downarrow
]

[
\boxed{
\text{Arithmetic Feature Reconstruction}
}
]

[
\downarrow
]

[
\boxed{
\text{Permutation / Null Controls}
}
]

[
\downarrow
]

[
\boxed{
\text{SFR + Rank Statistical Tests}
}
]

[
\downarrow
]

```
[
\boxed{
\text{Topological Tests}
}
]
```

```
[
\downarrow
]
```

```
[
\boxed{
\text{Cross-Dataset Replication}
}
]
```

```
[
\downarrow
]
```

```
[
\boxed{
\text{RTCH / Field / Energy Extensions}
}
]
```

This ordering prevents higher-level physical theories from being used to justify lower-level empirical claims.

49. Claim Readiness Matrix

Readiness	Claims
Ready to document	MAP-001, MAP-004, DATA-M001, DATA-M002, RTCH-M002, ECC-002
Ready for reconstruction	MATH-K001, MATH-K002, COSMO-L001, MAP-002, MAP-003, TOPO-001, SFR-E001, CTRL-RF001
Ready for controlled testing	SFR-A001, RANK-001, RANK-002, MAP-005, ARITH-001

Requires new theory/experiment	SFR-A002, FIELD-002, FIELD-003, ECC-003, ENERGY-001, COSMO-E001
Deferred/downstream	COSMO-E002
Historical only unless reconstructed	HIST-001 through HIST-008

50. Evidence Promotion Rules

An existing R&D artifact may be promoted from historical/exploratory evidence only when:

Mathematical

```
[
\text{derivation}
\rightarrow
\text{independent verification}.
]
```

Computational

```
[
\text{original code}
\rightarrow
\text{reproduction}
\rightarrow
\text{independent implementation}.
]
```

Statistical

```
[
\text{training result}
\rightarrow
\text{locked test}
\rightarrow
\text{null comparison}
\rightarrow
\text{replication}.
]
```

Physical

```
[
\text{mathematical consistency}
\rightarrow
\text{known limits}
\rightarrow
\text{existing constraints}
\rightarrow
\text{novel prediction}
\rightarrow
\text{independent observation}.
]
```

51. What the R&D Archive Has Already Achieved

The archive is considerably more valuable than a simple collection of preliminary experiments.

It already provides four major components of the S.T.A.R. research infrastructure:

1. Mathematical Candidate Layer

```
[
E\rightarrow I(E)\rightarrow\Phi(E)
]
```

is substantially developed.

2. Computational Candidate Layer

There are working implementations of:

- mappings;
- SFR models;
- ($L_{\rm cosmo}$);
- feature engineering;
- SHAP;
- statistical calculations;
- data matching.

3. Astronomical Data Layer

The project possesses substantial multi-survey data infrastructure.

4. Theoretical Extension Layer

RTCH/ECC provide candidate mathematical frameworks for the later physical-theory program.

What is still missing is the controlled bridge:

```
[
\boxed{
\text{candidate construction}
\rightrightarrows
\text{null-calibrated independent experiment}
}
```

52. Current Evidence Bottleneck

The most important conclusion from the crosswalk is that the principal bottleneck is **not lack of research material**.

The bottleneck is:

```
[
\boxed{
\text{reconstruction + controls + independent validation}
}
```

The R&D archive contains enough material to populate a substantial Claim/Evidence Registry.

It does not yet justify promoting the central cosmological claims to established evidence.

53. Recommended S.T.A.R. Evidence Architecture

The resulting research architecture should now be:

[
 \boxed{\text{Charter v0.2}}
]

↓

[
 \boxed{\text{Claim/Evidence Registry v0.2}}
]

↓

[
 \boxed{\text{Data Provenance Registry}}
]

↓

[
 \boxed{\text{Parameter Registry}}
]

↓

[
 \boxed{\text{Mapping Registry}}
]

↓

[
 \boxed{\text{Experiment Registry}}
]

↓

[
 \boxed{\text{Results Registry}}
]

↓

[
 \boxed{\text{Replication Registry}}
]

with the Claim → Evidence → Experiment relationship represented as:

```
[
\boxed{
C_i
\longrightarrow
E_{ij}
\longrightarrow
X_k
\longrightarrow
R_k
\longrightarrow
V_i
}
]
```

where:

- (C_i) = claim;
 - (E_{ij}) = evidence item;
 - (X_k) = registered experiment;
 - (R_k) = experimental result;
 - (V_i) = updated evidence/validation state.
-

54. Immediate Action Set

The crosswalk therefore reduces the next phase of S.T.A.R. to five concrete workstreams:

WORKSTREAM 1 — Reconstruct

```
[
K,\quad L_{\rm cosmo},\quad
\Phi,\quad
z=200r,\quad
\text{historical TDA}.
]
```

WORKSTREAM 2 — Clean

Establish:

- dataset lineage;

- matching parameters;
- leakage controls;
- object-level train/test separation;
- fixed preprocessing.

WORKSTREAM 3 — Establish Controls

Implement:

```
[
\text{permutation}
+
\text{random-feature}
+
\text{alternative invariant}
+
\text{alternative mapping}.
]
```

WORKSTREAM 4 — Run Priority-A Tests

Especially:

```
[
\text{rank/environment}
]
```

```
[
\text{rank/topology}
]
```

```
[
\text{arithmetic}\rightarrow\text{SFR}
]
```

```
[
\text{mapping comparison}.
]
```

WORKSTREAM 5 — Preserve the Negative Record

All failures, unstable (K) values, poor OOS performance, failed mappings, and null results remain in the registry.

They are not discarded.

55. Final Crosswalk Principle

The R&D archive should therefore be treated neither as proof nor as discarded preliminary work.

Its correct scientific status is:

$\text{Evidence Reservoir}$
}
]

from which evidence can be promoted only through:

[
 $\boxed{\text{Provenance}}$
 $\rightarrow$
 Reconstruction
 $\rightarrow$
 Control
 $\rightarrow$
 Experiment
 $\rightarrow$
 Replication
}
]

The central objective of v0.2 is consequently not to maximize the number of “supported” claims.

It is to make every claim answerable to a specific question:

What evidence supports this claim, where did that evidence come from, what experiment tests it, what null could defeat it, and what remains to be done before the evidence can be considered reproducible?

That is the controlling relationship for the S.T.A.R. research program.